CSA5108 Cryptography and Network security

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Exp 7 : Simple Substitution Cipher
Code:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main() {
   char cipher[] =
   "53‡‡†305))6*;4826)4‡.)4‡);806*;48†8¶60))85;;]8*;:‡*8†83"
   "(88)5*†;46(;88*96*?;8)*‡(;485);5*†2:*‡(;4956*2(5*—4)8¶8*"
   ";4069285);)6†8)4‡‡;1(‡9;48081;8:8‡1;48†85;4)485†528806*81"
   "(‡9;48;(88;4(‡?34;48)4‡;161;:188;‡?;";

   char key[256];
   char plain[5000];
   int i;


   for (i = 0; i < 256; i++)
      key[i] = '?';
   key['5'] = 'e';
   key['3'] = 't';
   key['‡'] = 'h';
   key['†'] = 'a';
   key['3'] = 't';

   printf("Ciphertext:\n%s\n\n", cipher);

   printf("Enter substitutions as pairs: cipher plaintext\n");
   printf("Enter 0 0 when finished.\n");

   while (1) {
```

```
        char c, p;

        printf("Mapping: ");
        scanf(" %c %c", &c, &p);

        if (c == '0' && p == '0')
            break;

        key[(unsigned char)c] = p;
    }

    for (i = 0; cipher[i] != '\0'; i++) {
        if (key[(unsigned char)cipher[i]] != '?')
            plain[i] = key[(unsigned char)cipher[i]];
        else
            plain[i] = cipher[i];
    }

    plain[i] = '\0';

    printf("\nDecrypted message:\n%s\n", plain);

    return 0;
}
```

